Decorating the Visuals Transcript

Welcome to BITS PLANET DIGITAL. Today we will continue our learning journey for the course Data Visualisation Storytelling. We will be now exploring some advanced features in the Bokeh environment while using the Python for building your visualisations.

As mentioned, Bokeh is a software within the Python environment for creating interactive and highly sophisticated visuals which improvise the look and feel of your data storytelling exercises. So, we will be now looking at how to improvise your plots or glyphs as they are called in Bokeh environment and try to bring in more pre-attentive attributes like the colour, labels, legend, and the styling of axes, so on and so forth. We will also try to look at how to apply some custom themes so that you can use them for improvising your graphics.

So, now let us start by running the code and understanding how these features work in the Bokeh environment. Like always, I will open a new notebook in the Colab environment and try to simultaneously run the code so that you can understand the functioning and attempt the same while practising using the Colab notebook that will be uploaded in the LMS page. So, for using Bokeh, you have to first import the output notebook from Bokeh and that is when you can apply the Bokeh commands and use them for your building the graphs and visualisations.

So, first I am importing the output notebook from Bokeh environment. The main objective of this lesson is to customise the appearance of your plots and this includes changing the title, axis labels, grid lines, and also how to add legends, okay, and some of the other custom themes, how to apply them, we will learn those features. So, colours in Bokeh.

So, defining colours works same across all different parts of Bokeh. There are different ways to define colours including the following. So, you have all the 140 HTML and CSS colours like you can directly call the name green and indigo or you can give an RGBA value using the hexadecimal format or you can use tuple of integers and also a four tuple of integers where the last RGBA indicates the transparency which is a floating point value between 0 to 1. Now, let us see how to use all these features one after another.

So, from the last lesson, we are using the same fruits, different types of fruits and the count of fruits and we are trying to use different colours for that. So, I am just copy pasting the code into the Colab book. So, here from Bokeh.plotting, I am importing figure and show.

So, remember you have to always show the figure in Bokeh as opposed to your SNS in Mathplotlib. In Bokeh, you have to use the show command for finally visualising the graphic. And what is the data that we are parsing? Fruits are apples, pears, nectarines, plums, grapes and strawberries and you have the count of fruits and you are using colour underscore plot.

Okay, we are defining the figure. Remember, we have to define the figure and then set the x-axis range which is a categorical variable. Remember, it is a categorical variable not a continuous variable.

So, by using exchange, you are telling the system that you have to use it as a categorical variable. Then height is set at 350. Okay, so then we are trying to use different colours and we are using the simple vertical bar.

So, now colour plot, we have defined the figure function and we are now calling the vertical bar function. So, we are just giving fruit and height is equal to count, top is equal to count, width colour is indigo. So, I am just running this part.

So, this is what you got. Okay, colour is equal to indigo. So, now you want to change the colour to different colour.

So, you can simply uncomment this part and as we run, it will take the most recent and you can look how the different colours are changing. Okay, so this is the way to define it. 0, 100, 100, it will give green colour.

This is the using of three tuples, which is the three tuple of integers indicating RGB. So, in this case, if you make this to be let us say 100, just testing it, don't know which colour it will come. It is coming grey or you can give it as 150, then this to be 10, this to be 10, it will give some other colour.

So, this is the red that is coming. And finally, what you can do, you can also use the fourth tuple number, the A parameter which sets the transparency and run the graph. So, you can see the difference.

Okay, so 100, 100, 100, maybe I will put 300. So, it will be grey and I am setting transparency to be. So, let us see first transparency is 85%.

I am running that. So, this is the graph you are getting. And now I set transparency to 15.

So, you can see the difference. So, you can change the A parameter to get the different transparency levels of the colours that you are using. You can also directly give the name of the colour, hexadecimal format or three tuples of colour.

So, what you have to remember here is, if you know a particular colour combinations RGB value or hexadecimal value, you can directly call in and use that for colouring your graph. So, then let us try to understand what are the visual properties that you can play around in the bokeh environment. So, now we will see how the following visual properties can be used in action.

Text font size, text colour, which are text properties, line width, line colour. Okay, line transparency, whether light type, whether it has to be dash that can come from line properties. Fill colour, fill alpha, fill transparency, hatch colour, hatch alpha, so on, which are hatch properties.

We will now see how these are used. And just for the broader knowledge purpose, you can always open the general visualisation guidelines. So, you can go to the source documentation within the bokeh documentation page and learn about more properties here.

So, again see how different types of things can be used, what is the abbreviation, what will it show and how to use them in the syntax. All these things are given here. There are certain examples.

Please practise them at your leisure, so that you will be thorough with how to use those properties. Now, let us try to import this code and see how we can use the different text formatting, line formatting, filling the colour formatting and hatch colour and transparencies. So, we are trying to show this.

So, let us run one segment after another segment. I am just copy pasting the code. So, what am I doing here? So, I will just explain the code and then run it.

I am importing pandas and importing numpy and also I am importing bokeh plotting. These are there. So, I am running the selection, run the selection.

Then I am setting up the figure. So, I am calling it as visual underscore properties underscore plot height and title I am giving here domestic freight mail, freight and mail. This shows how month wise the freight number of people, I mean the tonnes of cargo that has been transported and the mail that has been transported by an airline.

So, I am setting up the plot figure and calling it to the variable visual underscore properties underscore plot. Now, I am determining the data variables here. So, I am using data dictionary in which I am defining freight to be a random number between of size 12.

It is a tuple of size 12 ranging from 500 to 2000. Similarly, mail ranges from 100 to 500. So, now I am creating a data frame using the data dictionary and I am using pandas for this.

So, monthly values underscore df is defined after from the data dictionary using the pd dot data frame command. So, once I do this, I am getting a data frame. So, I am running the selection.

So, again I think I have run this. So, I run the selection. Now, I have the data frame.

I have the now figure plot everything. So, now I am determining x variable as take the data frame df monthly values df and use the index which is the number 12. Then freight is equal to monthly values subsided by freight mail monthly value subsided by mail.

So, we are defining this as x and y variables. Okay, x is x axis the months number of months that we have now the point here is we will try to. Okay, so we are trying to change the visual properties.

So, how can you use that? So, visual underscore properties is the figure function that you have determined after that you just put a dot and calling the title function and within the title function. This is for the title. What are you doing? You are trying to change the text font size to 1.2 em then same visual properties text colour to dark blue.

So, we this we will just try to do this too. Okay, so we are running the selection. So, this is change then we have trying to change the values of the okay.

So, we are trying to change the visual properties of the line same look at the logic you are calling the figure function for example, I could have defined it as ABC. So, then you will say this ABC dot line then you are determining it's a line plot with x as the x axis freight line width is 3 then you are leaving pass in the parameters for line width colour transparency and line type and you are running this. Okay, so actually we should get a run the selection.

Okay, so I am trying to so glyph object is created because the line plot is already created and I am trying to run the selection. I am getting the plot here. Okay.

So in fact it has taken the vertical bar also which is showing the number of with it is the mail. Okay, and also you have the freight and mail in the same graph and the hatch pattern is spiral. You have the yellow colour of the hatch alpha.

So the main takeaway from this particular code is you can define the figure plot a variable and you can use these specific elements that you want to customise by calling them afterwards like vertical plot visual properties underscore plot is a figure function then in that you want to change the title and within the title you want to change the text font like this. You can change so you got the light blue me. So what I am trying to see I will just trying to make it red here and re-run the entire code.

I am getting it in red and instead of 1.2 M. I am making it 1.80 M. So it becomes much larger. You can also use the location whether it is centre left aligned right aligned so on and so forth. Okay, so please experiment with these different styling options.

So that are customising options. So you are comfortable in customising in the graph in the way that you want. Now let us come to styling the plots.

You can define the appearance of many elements of a plot which includes the width and height outline border and background for example. So again this goes takes us back to the just let principles where you will be using the principles for improvising the look and feel of your graph to grab the attention of your audience. So now let us try to see how to customise or style a particular plot.

I have just copied the code taking back to the pull up book and here I am running the command. So I am defining a new figure here and it is simply plot is equal to figure and not passing anything into the figure. Please note the difference in earlier case while defining the figure.

I have determined the height. I gave the title here. Here I am just saying plot is equal to figure.

The figure function is passed into the variable plot. Now what I am doing. I am trying to assign different values to the plot.

There are two ways. One is to define everything within the figure. The other way is to determine you can subsequently after determining the figure variable or assigning the figure function to a variable.

You can call each one of the functions separately and customise them as you want. So now look at this plot height is set at 300 width is set at 700. All that you are doing is plot dot height plot dot width plot dot outline colour is navy outline colour is width is 2. Transparency is 0.5. You want to fill the background with light blue and you don't want grid line.

OK so X plot dot X grid grid line colour is none. So this is what you are trying to do. OK.

And then you are just plotting a scatter. This is very simple scatter. You directly pass at the variables 1 2 3 4 5 2 5 7 8 2 7 size is 10.

And finally you are trying to show the plot. So you got a simple plot like this. So you have OK.

This is what you have light blue. And in fact you don't want X grid line colour. Only Y grid is there.

X grid is not there. So I just make it grey. I should get a grey plot.

So like this you can customise. Note the difference. You need not pass everything in the figure.

You can pass it outside. OK. Then you can write about it.

Followed. Yeah. So it's very simple.

Please try out these examples so that you are thorough with how to style your graphs plots and glides. Now let us move to styling the glides. OK.

So you have already seen how to determine the properties of elements in your plot in our earlier classes. OK. So you can either set the properties of the glide when you define it for the first time like what we are doing earlier or you can set the properties of the glide after you have defined it to and you can access the visual properties of a glide after defining just by using the dot glide function.

OK. It's very easy. Like in the same case like the what we have used for figure glides also the same logic works.

Let us quickly see the examples. So I am just plotting a scatter plot. OK.

So I'm just plotting a scatter plot in the colab. So X comma Y. OK. Sorry.

I'm just copying the code and pasting it here. First we are defining the plot is equal to figure and in that I am determining height to be 300. So then what we are doing we are adding a circle glide and define its colour as yellow and red.

So we are just determining this plot should be a scatter plot and I am calling it as a circle to the variable is called circle and what are the data. So X is 1 2 3 Y is 2 5 8 size is 15 fill colour is yellow line colour is red. So this is what we have done.

So first let us. OK. So let us run only the.

I'll just comment this out so that you can know the difference. OK. I'm just commenting this out.

I am showing this. So this is what you have got. So look.

So it is simply setting the X axis and Y axis X is 1 2 3 Y is 2 5 8. We are getting these clear values fill colour is yellow and line colour is red line colour in this instance is the colour of the circle. OK. So now what we are trying afterwards you want to change the colour to green.

So circle. So the variable for which the scatter plot is drawn and you said dot glide dot fill colour is green. OK.

So we are just. So the colour is now changed to green and line colour you want to change to blue. You just do the circle dot glide dot line underscore colour.

So what you have to remember here is can directly pass it on the function or on the variable that you have determined the plotting function and then afterwards dot glide is the function you will use and dot fill colour dot fill line colour is a particular attribute that you want to change. So you can style your. We'll now see how to customise your axis.

So axis sometimes becomes a very important attribute that you want to customise in your graphs because you want to convey the attention onto the variable that you are measuring or the way that you are showcasing sometimes so that you will be able to clearly convey the quantitative information that you are getting across to your audience. So remember it is always advised that your axis represents the true zero and also conveys the rightful measurement of the quantities. So it might be advisable for you to represent it more clearly.

So the information that you are presenting is transparent and easily understood by your audience. So like in the case of line you can also modify the colour thickness and grid lines and use the x and y axis attributes for configuring your axis. Now let us quickly look at this example and then you can understand how to use these properties.

So again I am copying the code and pasting it in the Colab environment. So again I am defining the figure function to a variable called axis underscore plot and on this axis plot axis plot I am trying to plot a line graph by saying axis underscore axis plot dot line which will plot a line function with x and y values and line colour is purple. So this is what I have done initially.

So it is selecting everything. So let me just comment them out so that you can see the difference. So I am just commenting this out.

So you can know the difference. So this is the initial thing. So we are just getting the purple line for x and y values which is a simple line graph.

We just got the values for x and y and you have the purple line. So in fact what you can do I am increasing it to 5 to 12. So just for fun.

So I have increased it to this. Now this is how your graph is looking like. Now you want to customise it.

So first you want to add line colour to x axis. So you just use the axis underscore plot dot x axis and axis line colour is blue. So you can colour your x axis by using this function as blue or red.

Similarly you want to change the minor declines which are the smaller declines that you have. You want to colour them as orange. So we are doing that.

So you can see they have become orange and you want to change the major decline to be red. So you can just call the major declines. What you are simply doing is the syntax is you are calling the function and determining what you want to modify.

I want to modify x axis and what is that you want to modify in x axis dot after the dot you are writing axis underscore line underscore colour minor underscore tick underscore line colour right like that you can customise so many things. So we have modified that and we have made it red colour. Now we are going to the y x axis major label.

You want to increase the font size of the major label OK which is seemingly small 1 2 3. In fact I would also make it 1.8 would look at large make it large. So you have got to increase the size. Then also you want to change the y axis line colour and do the same thing for y axis.

This is how you can customise your axis in the bokeh environment. Sometimes you want to change the orientation of labels that are on your x and y axis. For example you have a name of a month which is horizontal.

It might be useful to turn it into vertical so that the readability increases or the space looks more elegant. Now let us try to understand how to use the features in bokeh to achieve the orientation of the major labels on the x and y axis. So you can also go and see the documentation tick label orientation tick label formats.

Now let us quickly look at a example for doing this. So for that you need this numerical tick formatter function which you will be importing first. So I'm just importing the numerical tick formatter function and pandas and numpy for doing the necessary tasks.

So I have different airways numbers and I have the passenger count. OK. And for each passenger the each aircraft you'll have the number of passengers.

So then I'm creating a data frame. So all these things by now should be very familiar to you. So you can again go through the code just running them for getting our data or focus is on creating the graph then or more about the data frame.

So what I'm doing I'm first trying to define the figure. I'm plotting a vertical bar with the following properties. So let me first do this then I will show how the ticker works.

So it is nothing. I'm defining a figure. X is a categorical variable which is unique carrier name and right.

And then the title is top 10 carriers by passengers and we are limiting it to top 10 only height and width I have defined. And then I'm trying to plot the vertical bar X. X is again a variable name of the particular airline and the length is given by the number of passengers subsided by the data frame by calling out the particular passengers column and legend label is passengers. Then you have the width.

OK. So we are the life is created and I have to use the show function to understand what kind of graph has been generated. So I've done this.

So this is the top 10 carriers. So you see all of them are cluttered and I'm not able to read this. So ideally it should be vertical.

So that are some some slanting position so I can read it clearly. And also Y axis variable is very large in value. I'm not able to make any sense of it.

It looks very clumsy. So I'm using these two properties. So this is the figure right.

Carriers by passenger underscore plot is the figure function that I have determined. So carriers by passengers and plot. Then to that I am calling the X axis.

I have called the X axis then underscore major label orientation. This is the function and I am saying 0.8. Similarly for Y axis formatter I have just used the formatter where numerical formatter and says the format should be 0. There should be rounded to 0. So this is what I have done. So again I have to sorry I have to use them within the before the show function so that the life will be created.

So I'm just using this modifications took life. So now you see these two graphs. So it is 0.8 is clearly aligned and the numbers are also instead of 1.4 into 10 to the power of 7 it is completely given the entire numerical and rounded to 0 digits.

So now if I give 2 here OK and make it 0.6 this is how this will look like. OK. So there are.

Thank you. So this is what we can. So this is how we can customise the axis in the bouquet environment.

So now let us look at the customising titles. OK. So now let us look at customising.

So titles are very important because they are the ones which will tell what is the graph representing and mostly people will immediately look at the title to understand the elements that are there in the graph. So one way that we have been using here is to determine your title within the figure function when you are first determining it or you can also first determine the basic figure and then add title by simply calling out the function plot dot title dot text and you can give your title. So now let us try to work out an example so that it will be very clear.

So I'm just copying it here and then taking it to the Colab environment. So I'm just pasted here from bouquet. OK.

These are already there. X and Y variables are determined. OK.

So and then you have the figure function which is determined where I am defining headline example which is the title and height I have set. And after that I am plotting a line function with X comma Y with line width is equal to 2. These are the basic things that I am doing. And then first let us try to uncomment all of this.

OK. And we will then see how it will impact your title. OK.

We'll uncomment each one of these and then see how your title can be customised. So simply headline example I have plotted the graph and then I gave the headline title with the title within the figure function and these are X and Y values. So very simple plot.

Now what I'm doing first I am trying to change the title location to left. OK. So I have changed the title location to left.

It was on top. I shifted to left. Instead of that had I given right here.

OK. It would come on this side. OK.

Then I'll just take it to the top again. So you should not use top. You should be using above for fixing this.

So I'm just modifying it as above. So it is back on there. Then I am just trying to change the title as updated plot title.

So again I am running that. So you have this. Then I am also changing it to font size to 25 percent and I am aligning it to right.

OK. And then background colour is dark grey and also title colour to white. So in fact white would not be visible here.

The updated plot title with background. Yes. So this is how we have now changed the title properties by calling out different functions.

Simple logic like always first call out the figure function titles underscore plot which is the figure function that we have determined then title location title text title font size alignment background. All these are directly you are calling the functions and you are able to change the features of the title. Now let us look at how to customise legends.

So you must have noticed by now already when you use the book a software it is automatically adding legend to your plot if you use the legend underscore label function. And now we will try to understand how to modify the legend location whether you want at the top bottom centre right. What is the legend title how to customise all of the features that are there in the legend.

And as I said earlier you can learn the same thing from the visual properties. So now let us prepare some data and use to see and see how legend properties work. So we are just copy pasting.

I mean we are just creating an x axis variable and two y variables y 1 and y 2 and determining figure function to a variable called legends underscore plot. A title is passed within that figure function legend example. Then you want to determine a line function with legend underscore plot the figure function and you are calling the line variant line imposing the line function so that x and y 1 will be created legend label is temperature line colour is blue line width is 2. Then you are doing the circle legend plot underscore scatter you are using x and y to determine the scatter plot x and y to have the scatter plot again you have determined the other attributes here.

Now what we are trying to do is we are just trying to create a simple graph here first we do the graph part then come to customising the legend. OK then you also need to show this plot. So let us try to first see the initial graph.

How does it look like then talk about the other aspects. So this is how the graph look like. So you have the blue line and the circles representing the objects and the temperature.

OK so these are your x and y attributes. Now we want to customise how does this particular legend looks like. So we have the legend blue line indicating temperature circle indicating the object.

We want to customise both of them. So I'm just trying to do all the following. The first I want to legend location to be top left.

Right now it is top right. I want to change it to top left. So then legend legend title should be observations.

It should come on the title of the legend box should be observations. OK legend label text colour should be navy. Time should font should be Times Roman font style should be italic.

There should be italic then background of the legend. You want to have the borderline with the three colour of navy transparency 80 percent fill also navy with transparency of 20 percent. So this is how you are getting this.

So in fact if you increase the transparency to 80 also here and make the legend borderline transparency to be less. So this is how it will look like. OK so you can use these features to again customise your legend.

Again the logic remains simple. You call the figure function variable. Append the variable or the attribute that you want to change and what feature of that attribute you want to change and then give these specific entries.

You will achieve the customisation. It's pretty straightforward. So as we have seen in Seaborn and in the math plot you can use the different colour mappers of pellets which are predefined for mapping colours onto your data which sometimes can be very useful especially for drawing the heat maps are showing a distribution of objects.

These colour pellets are very useful. So you have this predefined bokeh underscore bokeh dot pellets function which is there or module which is there in the bokeh environment can import it and you can use this for colouring your data. So let us try one simple example.

First we have to import the colour pellets. One of such pellets is CIDES. This is a colour pellet which has the following hexadecimal codes which are there in that particular module.

So our particular pellet the first 10 colours look like this. This is what we are trying to do. So we are just importing the particular colour pellet.

Then we have imported it. We have seen what colours are there in that. Then we can also use the different types of colour mappers.

One is a linear colour map which maps colour ranges according to linear which are mapped linearly. Then also you have a logarithmic C map which colours based on a based on logarithmic values. It colours logarithmically.

A linear colour would have a same gradient. Logarithmic colouring would have an increasing gradient. So let us see how to use this for on a particular data set.

I am just importing the show function, a figure and you have to import the linear C map function from the transform module of the bokeh. And x is minus 32 to plus 33. Then y is nothing but x square.

This is a simple parabola and then you are determining the figure function calling it as mapper underscore plot. So you are just trying to do a scatter plot by mapper underscore plot is the variable. You are calling the scatter function passing x and y value and colour you are giving linear C map.

For colour you are giving linear C map based on the y value and the turbo 56 is the particular colour pellet you are using. Boundary conditions are min and max of y size is 10 and you are trying to show the plot. So this is how your plot looks like and the colours are given.

The colours are given in this. A pellet is used where the bottom most values have singular colour. Then the extreme values have a different colour.

The range of colours are used. It is a parabola coloured differently or rather colours are mapped using a linear C map and the pellet uses turbo 256. Now the question comes is what about the legend and which colour represents what numerical value to this end.

You can add the simple function called colour bar which will be added to the graph onto the right side or left side depending upon how you choose to place it. And it can be useful for assigning a meaning to the different colours that are there in the graphic. So by just adding this colour underscore bar so what we are doing we are trying to add this function using the function construct colour bar.

So construct colour bar tries to give you the colour bar next to your graphic and you can simply add it to the plot. Okay. Mapper plot add layout and we are trying to use the colour object and you want to place it right.

I can also place it on the left side. Okay. I can place it on the left side.

So if you choose to do so then simply based on the left side. So you can achieve these functions using the construct underscore colour bar function for adding the colour bars to your graphic which will then correspond to different values and that can be easily deciphered by your graph. Audience.

So like in matplotlib and seaborn we also have specific themes that are predetermined visualisations styling patterns that are there in the book. So there are many such themes which can which you can directly use with preset attributes and colouring schemes so that you need not worry about customisation at large. So you can use a predetermined scheme and then improve it from there as as and referred for your particular case and you can apply this theme to the whole document.

Okay. And the whole document is the output document that notebook that we are talking about where the different graphs and plots are there. So different widgets everything can be set into a very nice predetermined theme.

So it looks continuous for your audience. Now let us try to see one of such functions. So and how we can do the usage of themes.

So let us copy paste the code and understand the code. So first we are trying to import the current doc or the current document function. So this is one which will allow you to apply the theme for the current object that you are trying to work on.

So. Okay. So first I'm trying to set current document theme calibre.

Okay. Then. Okay.

First I'm using the night sky theme. It is nothing but the X and Y plot a simple line plot between X and Y. And I'm showing the document. So this is though it is the night sky which actually looks darker.

So I'm not liking the night sky because it is dark in the back. I'm just now trying to run the calibre theme which is very simple and straightforward pure colour white. Then I'm trying dark minimal.

Okay. Again this is dark minimal. Then also you have the light minimal.

So which is like this. Then you have the contrast. These are the preset themes which you can use for achieving these stylised themes in the bokeh environment and you can set them at the beginning and subsequently everything will come in the same thematic format giving some continuity to the entire visualisations that you are trying to generate.

Okay. Thank you.

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